

PROJECT

CS-844: GENERATIVE DEEP MODELS

**PAPER REVIEW: NVAE: A DEEP HIERARCHICAL
VARIATIONAL AUTOENCODER**

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1. Comprehensive Summary

NVAE (Nouveau VAE) by Vahdat and Kautz, published at NeurIPS 2020, presents a carefully designed deep hierarchical Variational Autoencoder (VAE) that achieves state-of-the-art image generation quality among likelihood-based generative models at the time of publication. The central thesis of the paper is that the historically poor image quality of VAEs relative to GANs is not an inherent limitation of the VAE framework itself, but rather a consequence of inadequate model capacity and training instability when scaling VAEs to greater depth and resolution. NVAE demonstrates that by combining a principled hierarchical latent variable structure with targeted architectural innovations to promote training stability, VAEs can generate images of dramatically higher visual quality than previously demonstrated.

The model adopts a top-down hierarchical architecture in which latent variables are organized in L groups, with each group operating at a specific spatial resolution. Unlike bottom-up (ladder) VAEs, NVAE uses a bidirectional top-down inference model: the encoder (bottom-up path) computes approximate posterior distributions at each level, while the decoder (top-down path) defines the prior for each latent group conditioned on the groups above it. The key parameterization for this hierarchical structure uses residual normal distributions, where the prior at each level is not a fixed standard Gaussian but is instead a Gaussian whose mean and variance are predicted by the decoder from the higher-level latent variables. This allows fine-grained information to be stored in later (higher-resolution) groups, while coarse global structure is captured in the top-level groups.

The paper makes several critical architectural and training contributions that enable this depth to be achieved without training instability. First, the generative model cells use depthwise separable convolutions combined with standard convolution layers organized in residual blocks with batch normalization and Squeeze-and-Excitation (SE) modules. The use of SE modules allows the model to adaptively recalibrate channel-wise feature responses, which is crucial for capturing long-range spatial correlations within each resolution group. Second, spectral normalization is applied to the convolutional weights throughout the model to bound the Lipschitz constant and prevent exploding activations during the early phases of training. Third, a modified batch normalization strategy with gradient skipping is introduced: the running statistics accumulated in batch normalization layers during the forward pass are frozen and gradients are not propagated through them, preventing the training instability that arises when large gradients perturb batch statistics. Finally, the Kullback-Leibler (KL) divergence terms in the ELBO are balanced using a KL balancing technique that prevents posterior collapse—a common failure mode in deep VAEs where the model ignores the latent variables.

NVAE achieves bits-per-dimension (BPD) scores of 2.91 on CIFAR-10, 3.92 on ImageNet 32x32, and 4.53 on ImageNet 64x64 in unconditional generation, significantly outperforming all prior VAE-based methods. On CelebA-HQ 256x256, NVAE generates visually sharp portrait images that, while not matching the fine-grained detail of style-based GANs, are far superior to any previous VAE. The paper also demonstrates competitive FID scores and provides a thorough ablation study establishing the contribution of each architectural component.

2. Type of Generative Deep Model

NVAE is a Variational Autoencoder (VAE), specifically a Deep Hierarchical Latent Variable Model (HLVM). VAEs are likelihood-based generative models that define a joint distribution $p_{\theta}(x, z)$ over observed data x and latent variables z , with the generative model (decoder) factored as $p_{\theta}(x, z) = p_{\theta}(x|z) p(z)$. Training maximizes the Evidence Lower Bound (ELBO):

$$ELBO = E_{q[\sum \log p_{\theta}(z_l | z_{<l})]} - D_{KL}(q\phi(z|x) || p_{\theta}(z))$$

In NVAE's hierarchical formulation, the latent space is structured into L groups $z = \{z_1, z_2, \dots, z_L\}$ processed in a top-down order. The joint prior is therefore:

$$p_{\theta}(z) = p(z_1) \cdot \prod_{l>1} p_{\theta}(z_l | z_{<l})$$

This hierarchical prior enables the model to represent multi-scale structure in the data a latent structure that flat VAEs with a single Gaussian prior cannot capture. NVAE can be seen as occupying a middle ground between VAEs and autoregressive models: it models long-range dependencies through the hierarchy of latent groups rather than through explicit pixel-level autoregression.

3. Advantages Over Prior and Baseline Methods

Superior image quality over standard VAEs. The original VAE and its shallow variants (beta-VAE, WAE) generate blurry, low-resolution images because a single-level Gaussian prior cannot capture the complex multimodal structure of natural image distributions. NVAE's hierarchical prior, operating at multiple spatial resolutions, allows the model to represent fine-grained local textures in higher groups while capturing global structure in lower groups, enabling sharp and diverse samples.

Outperforms prior hierarchical VAEs. Earlier hierarchical VAEs such as Ladder VAE (LVAE), BIVA, and Very Deep VAEs (VDVAE) suffered from training instability and posterior collapse that limited their depth. NVAE's combination of spectral normalization, modified batch norm, SE modules, and KL balancing directly addresses

these failure modes and enables a substantially deeper and wider model than its predecessors, yielding superior BPD scores on all evaluated benchmarks.

Tractable likelihood. Unlike GANs, NVAE directly optimizes a lower bound on the data log-likelihood, providing a principled probabilistic framework for evaluating model quality. This allows meaningful comparison across models using BPD metrics and makes NVAE suitable for downstream tasks such as density estimation, anomaly detection, and semi-supervised learning tasks for which GAN-based models are ill-suited.

Stable training at scale. Scaling standard VAEs to very deep architectures results in gradient instabilities and posterior collapse. NVAE demonstrates that with the right set of architectural choices particularly spectral normalization, gradient-decoupled batch normalization, and residual normal distributions it is possible to train very deep hierarchical VAEs reliably at high resolutions, a feat that was previously elusive.

Single-network end-to-end training. Unlike GANs, which require training two competing networks simultaneously and navigating a complex equilibrium, NVAE trains a single encoder-decoder pair using a well-defined objective (the ELBO). This simplifies the training setup and removes the mode collapse and training instability characteristic of adversarial methods.

4. Downsides and Limitations

- Memory and compute intensity. NVAE is one of the most computationally demanding VAE architectures ever proposed. The combination of many hierarchical groups, large residual blocks with SE modules, multi-scale feature maps, and spectral normalization requires substantial GPU memory. Training NVAE at 256x256 resolution requires multiple high-memory GPUs and multi-day training runs, making it inaccessible to researchers without significant compute budgets.
- Still inferior to GANs on perceptual quality metrics. Despite the significant improvements over prior VAEs, NVAE's FID scores and visual sample quality on high-resolution benchmarks remain notably below those of StyleGAN2 and similar GAN architectures. The VAE framework, constrained by the ELBO, tends to slightly blur fine-grained details even with a very deep hierarchy, because the pixel-level reconstruction likelihood encourages averaging across modes of the posterior.
- KL vanishing risk. Despite KL balancing, posterior collapse where individual latent groups are ignored because the ELBO is minimized primarily through the

reconstruction term remains a concern in very deep hierarchical VAEs. The KL balancing technique mitigates this but does not fully eliminate it, and careful hyperparameter tuning is required to maintain active latent groups at all hierarchy levels.

- Slow sampling relative to single-pass GANs. Although NVAE's sampling is much faster than DDPM (it requires only a single top-down pass through the decoder), each sampling step involves generating latent variables group by group in a sequential top-down fashion. This means the effective computational graph at inference time is sequentially dependent and cannot be trivially parallelized across groups.
- Architecture complexity and reproduction difficulty. NVAE introduces a large number of interacting design choices SE modules, spectral normalization, modified batch norm, residual normal distributions, KL balancing, and warmup schedules. Reproducing or extending NVAE requires careful attention to all of these components, as ablations show that removing any one of them significantly degrades performance.

5. Subsequent Improvements and Future Directions

NVAE demonstrated convincingly that VAEs have far greater expressive capacity than previously realized, inspiring a wave of subsequent work.

Very Deep VAEs (VDVAE). Child et al. (ICLR 2021) scaled up the hierarchical VAE paradigm even further, training models with hundreds of latent groups and achieving improved BPD on CIFAR-10 and ImageNet. VDVAE simplified some of NVAE's architecture while pushing depth and width further, demonstrating that the hierarchical VAE framework scales well with additional capacity.

Latent Score-based Generative Models (LSGM). Vahdat et al. (NeurIPS 2021) a direct follow-up from the same group combined NVAE's hierarchical VAE encoder with a score-based (diffusion) prior in the latent space. This hybrid approach achieves the generative power of score-based models in a much lower-dimensional latent space, yielding superior FID scores to both vanilla DDPM and NVAE while being more computationally tractable than pixel-space diffusion.

Diffusion-VAE hybrids. The broader research direction of combining VAE encoders (for compression and structure) with diffusion decoders (for sharp, high-fidelity generation) has yielded a productive family of models including Stable Diffusion, which uses a VAE to compress images into a latent space before running a diffusion process there. NVAE's demonstration that VAEs can produce meaningful structured latent representations at scale was an important precursor to this line of work.

Posterior collapse mitigation. Subsequent work on posterior collapse, including δ -VAE, Lagging Inference Networks, and Free Bits, has provided principled alternatives to KL balancing. Combining these with NVAE's architecture could further improve latent space utilization without the need for careful manual tuning of the KL balancing hyperparameter.

Disentanglement and controllability. NVAE's hierarchical latent structure is a natural candidate for disentangled representation learning. Future work could incorporate disentanglement objectives (such as those in beta-TC-VAE or FactorVAE) to produce interpretable latent groups where different groups control different semantic attributes of the generated image, improving controllability for image editing applications.